



UNIVERSITY
OF WOLLONGONG
IN DUBAI

MMMB215 Project Submission 2

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Subject number and name:	MMMB215 – Mechanical Design 1
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Title of Assignment:	Project Submission 2
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Lab Number:	7
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Student declaration and acknowledgement

By submitting this assignment online, the submitting student declares on behalf of the team that:

1. All team members have read the subject outline for this subject, and this assessment item meets the requirements of the subject detailed therein.
2. This assessment is entirely our own work, except where we have included fully documented references to the work of others. The material contained in this assessment item has not previously been submitted for assessment.
3. Acknowledgement of source information is in accordance with the guidelines or referencing style specified in the subject outline.
4. All team members are aware of the late submission policy and penalty.
5. The submitting student undertakes to communicate all feedback with the other team members.

Design Tasks

1.1. Task 1: Shaft Loading/Stress Analysis

Analytical analysis

In submission 1, the torques calculated were the **theoretical required** torques needed in order to be able to lift the 1 kg weight. They are now readjusted to the **actual** torque generated based on the 2.1 W power provided by the motor at an operational speed of 11,000 RPM. To do this, we find out the system's overall operating power or efficiency by dividing the actual power used by 2.1 W to evaluate the efficiency profile.

1. Actual Motor Input Torque

$$\omega_{motor} = \frac{11,000 \times 2\pi}{60} = 1,151.92 \text{ rad/s}$$

$$\tau_{motor} = \frac{Power}{\omega_{motor}} = \frac{2.1 \text{ W}}{1,151.92 \frac{\text{rad}}{\text{s}}} = 0.001823 \text{ Nm}$$

New gearbox ratios

Stage 1 (36/12) = 3.0

Stage 2 (48/12) = 4.0

Stage 3 (60/12) = 5.0

Total system gear ratio : 3 x 4 x 5 = **60**

2. Actual Torque

We assume 100% efficiency in our calculations.

$$\tau_A = \tau_{motor} \times GR_1$$

$$\tau_B = \tau_A \times GR_2$$

$$\tau_{output} = \tau_B \times GR_3$$

System Component	Speed (rpm)	Actual Torque (Nm)
Motor Input Shaft	11,000	0.001823
Shaft A	3,666.67	0.005469
Shaft B	916.67	0.021876
Output Shaft	183.33	0.109380

Force Analysis Diagrams

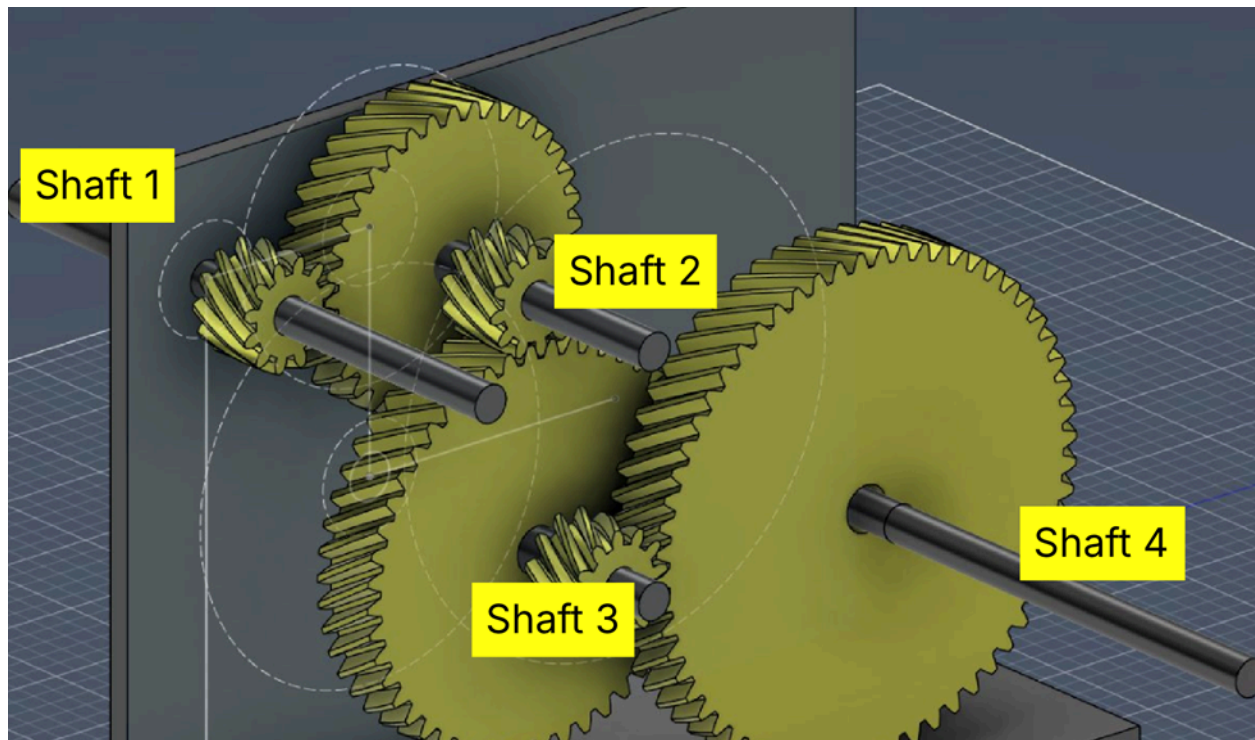
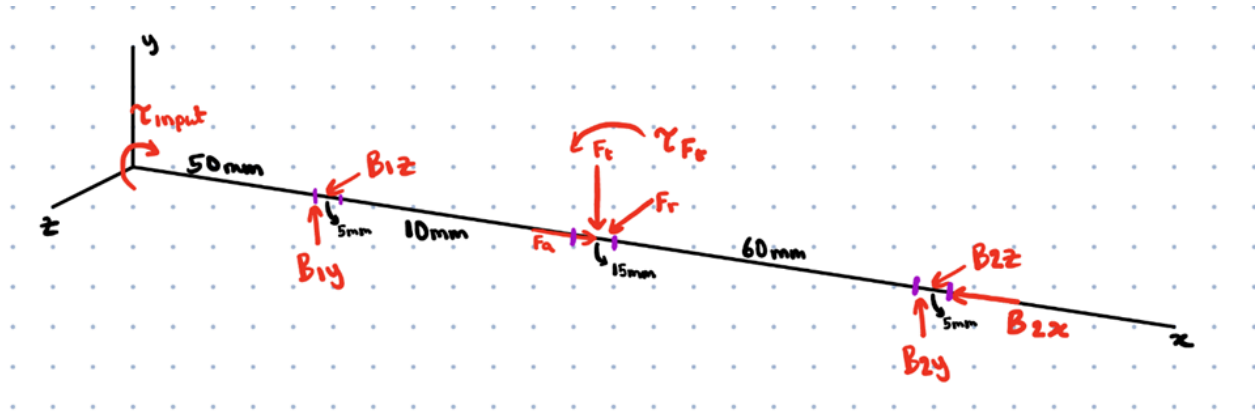


Figure 1: Gear and Shaft Layout

SHAFT 1:

N.B: (Torque/ pinion radius) was not used to find the tangential force, however it yields the same results as if it was calculated using that formula.



yz plane:

$$\sum M_O = 0 = -\tau_{\text{input}} + \tau_{F_t} = -\frac{60H}{2\pi n} + \frac{F_t d}{2} = -\frac{60(2.1)}{2\pi(11000)} + \frac{F_t((23.785 - 3) \times 10^{-3})}{2}$$

$$\rightarrow F_t = 0.175 \text{ N}$$

$$F_r = 0.175 \tan \phi_t = 0.175 \frac{\tan \phi_n}{\cos \psi} = 0.175 \frac{\tan 20}{\cos 30} \rightarrow F_r = 0.07 \text{ N}$$

$$F_a = 0.175 \tan \psi = 0.175 \tan 30 \rightarrow F_a = 0.1 \text{ N}$$

xy plane:

$$\sum M_{B1} = 0 = -F_t \left(\frac{5}{2} + 10 + \frac{15}{2} \right) + B_{2y} \left(\frac{5}{2} + 10 + 15 + 60 + \frac{5}{2} \right) = -0.175(20) + B_{2y}(90)$$

$$\rightarrow B_{2y} = 0.04 \text{ N}$$

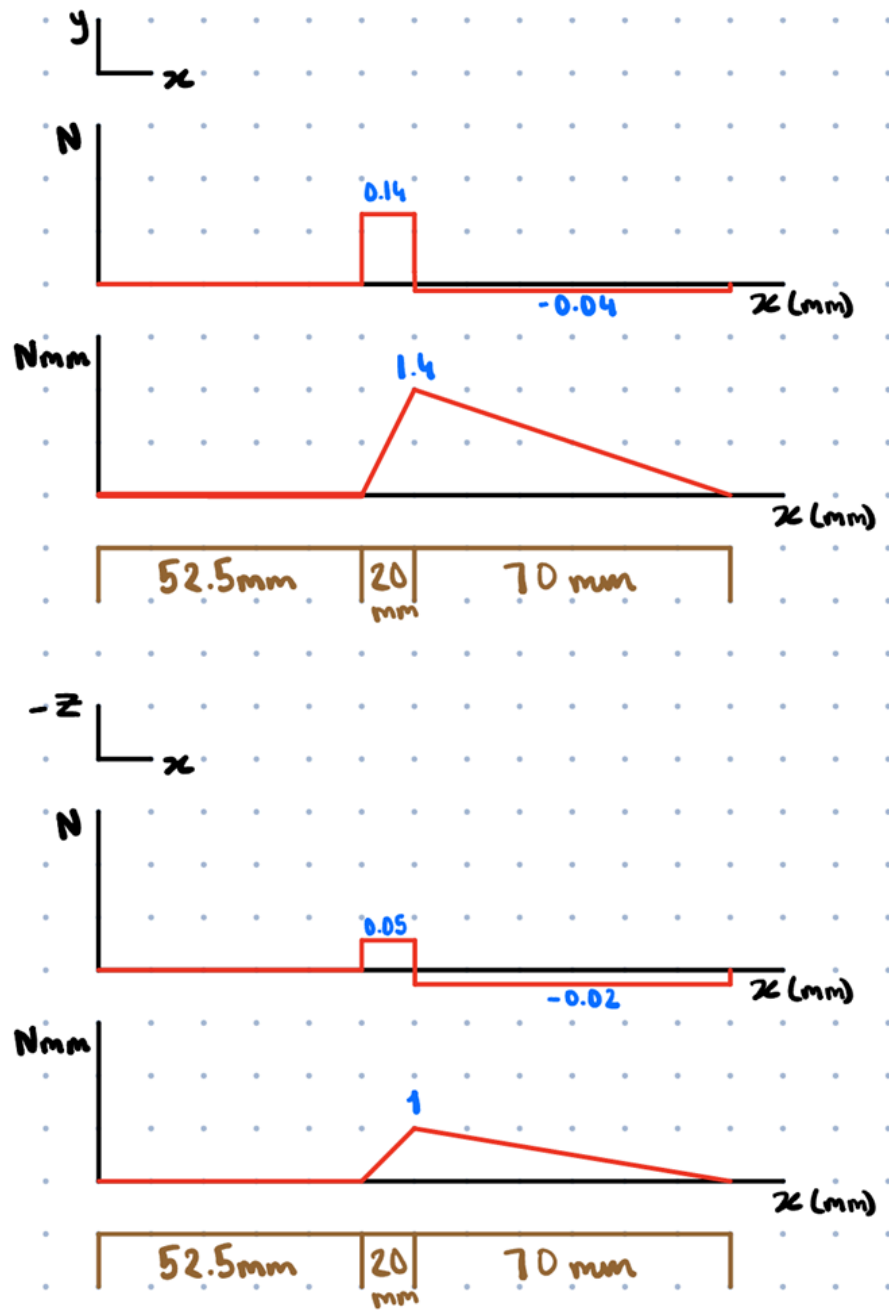
$$\sum F_y = 0 = B_{1y} + B_{2y} - F_t = B_{1y} + 0.04 - 0.175 \rightarrow B_{1y} = 0.14 \text{ N}$$

$$\sum F_x = 0 = -B_{2x} + F_a = -B_{2x} + 0.1 \rightarrow B_{2x} = 0.1 \text{ N}$$

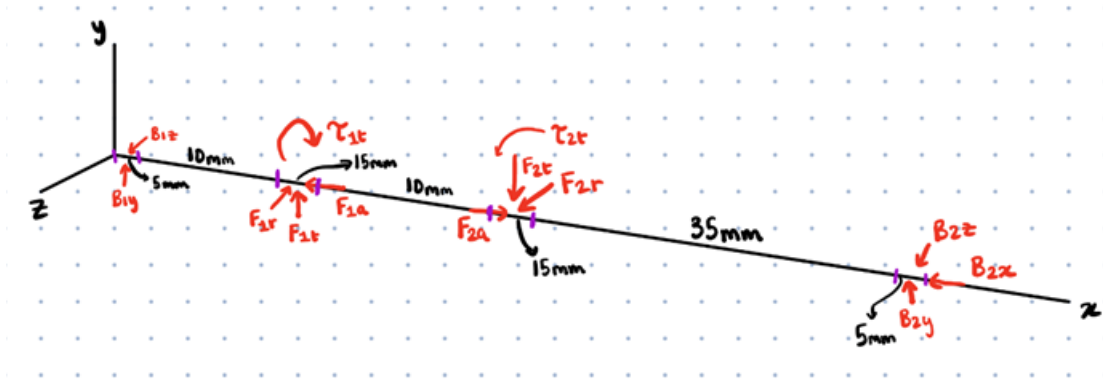
xz plane:

$$\sum M_{B1} = 0 = -F_r \left(\frac{5}{2} + 10 + \frac{15}{2} \right) - B_{2z} \left(\frac{5}{2} + 10 + 15 + 60 + \frac{5}{2} \right) = -0.07(20) - B_{2z}(90) \\ \rightarrow B_{2z} = -0.02 \text{ N}$$

$$\sum F_z = 0 = B_{1z} + B_{2z} + F_r = B_{1z} - 0.02 + 0.07 \rightarrow B_{1z} = -0.05 \text{ N}$$



SHAFT 2:



From shaft 1 calculations:

$$F_{1r} = 0.07 \text{ N}$$

$$F_{1t} = 0.175 \text{ N}$$

$$F_{1a} = 0.1 \text{ N}$$

yz plane:

$$\begin{aligned} \sum M_O = 0 &= -\tau_{1t} + \tau_{2t} = -\frac{60(2.1)}{2\pi(11000)}(3) + \frac{F_{2t}d}{2} \\ &= -\frac{60(2.1)}{2\pi(11000)}(3) + \frac{F_{2t}((23.785 - 3) \times 10^{-3})}{2} \rightarrow F_{2t} = 0.526 \text{ N} \end{aligned}$$

$$F_{2r} = 0.526 \tan \phi_t = 0.526 \frac{\tan \phi_n}{\cos \psi} = 0.526 \frac{\tan 20}{\cos 30} \rightarrow F_{2r} = 0.22 \text{ N}$$

$$F_{2a} = 0.526 \tan \psi = 0.526 \tan 30 \rightarrow F_{2a} = 0.3 \text{ N}$$

xy plane:

$$\begin{aligned} \sum M_{B1} = 0 &= F_{1t} \left(\frac{5}{2} + 10 + \frac{15}{2} \right) - F_{2t} \left(\frac{5}{2} + 10 + 15 + 10 + \frac{15}{2} \right) \\ &\quad + B_{2y} \left(\frac{5}{2} + 10 + 15 + 10 + 15 + 35 + \frac{5}{2} \right) \\ &= 0.175(20) - 0.526(45) + B_{2y}(90) \rightarrow B_{2y} = 0.22 \text{ N} \end{aligned}$$

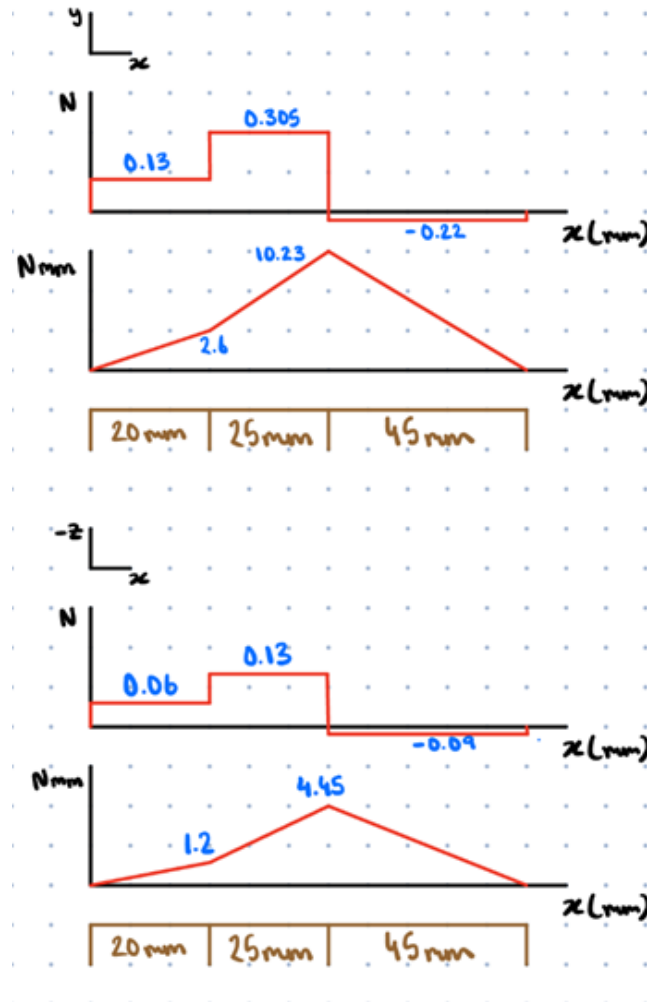
$$\sum F_y = 0 = B_{1y} + B_{2y} + F_{1t} - F_{2t} = B_{1y} + 0.22 + 0.175 - 0.526 \rightarrow B_{1y} = 0.13 \text{ N}$$

$$\sum F_x = 0 = -B_{2x} - F_{1a} + F_{2a} = -B_{2x} - 0.1 + 0.3 \rightarrow B_{2x} = 0.2 \text{ N}$$

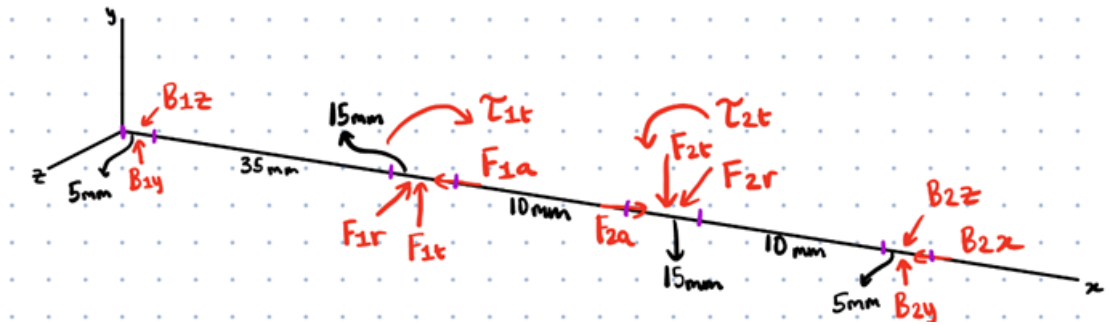
xz plane:

$$\begin{aligned} \sum M_{B1} = 0 = F_{1r} \left(\frac{5}{2} + 10 + \frac{15}{2} \right) - F_{2r} \left(\frac{5}{2} + 10 + 15 + 10 + \frac{15}{2} \right) - B_{2z} \left(\frac{5}{2} + 10 + 15 + 10 \right. \\ \left. + 15 + 35 + \frac{5}{2} \right) = 0.07(20) - 0.22(45) - B_{2z}(90) \rightarrow B_{2z} = -0.09 \text{ N} \end{aligned}$$

$$\sum F_z = 0 = B_{1z} + B_{2z} - F_{1r} + F_{2r} = B_{1z} - 0.09 - 0.07 + 0.22 \rightarrow B_{1z} = -0.06 \text{ N}$$



SHAFT 3



From shaft 2 calculations:

$$F_{1r} = 0.22 \text{ N}$$

$$F_{1t} = 0.526 \text{ N}$$

$$F_{1a} = 0.3 \text{ N}$$

yz plane:

$$\begin{aligned} \sum M_O = 0 &= -\tau_{1t} + \tau_{2t} = -\frac{60(2.1)}{2\pi(11000)}(3 \times 4) + \frac{F_{2t}d}{2} \\ &= -\frac{60(2.1)}{2\pi(11000)}(12) + \frac{F_{2t}((23.785 - 3) \times 10^{-3})}{2} \rightarrow F_{2t} = 2.1 \text{ N} \end{aligned}$$

$$F_{2r} = 2.1 \tan \phi_t = 2.1 \frac{\tan \phi_n}{\cos \psi} = 2.1 \frac{\tan 20}{\cos 30} \rightarrow F_{2r} = 0.88 \text{ N}$$

$$F_{2a} = 2.1 \tan \psi = 2.1 \tan 30 \rightarrow F_{2a} = 1.2 \text{ N}$$

xy plane:

$$\begin{aligned} \sum M_{B1} = 0 &= F_{1t} \left(\frac{5}{2} + 35 + \frac{15}{2} \right) - F_{2t} \left(\frac{5}{2} + 35 + 15 + 10 + \frac{15}{2} \right) \\ &+ B_{2y} \left(\frac{5}{2} + 35 + 15 + 10 + 15 + 10 + \frac{5}{2} \right) = 0.526(45) - 2.1(70) + B_{2y}(90) \\ &\rightarrow B_{2y} = 1.37 \text{ N} \end{aligned}$$

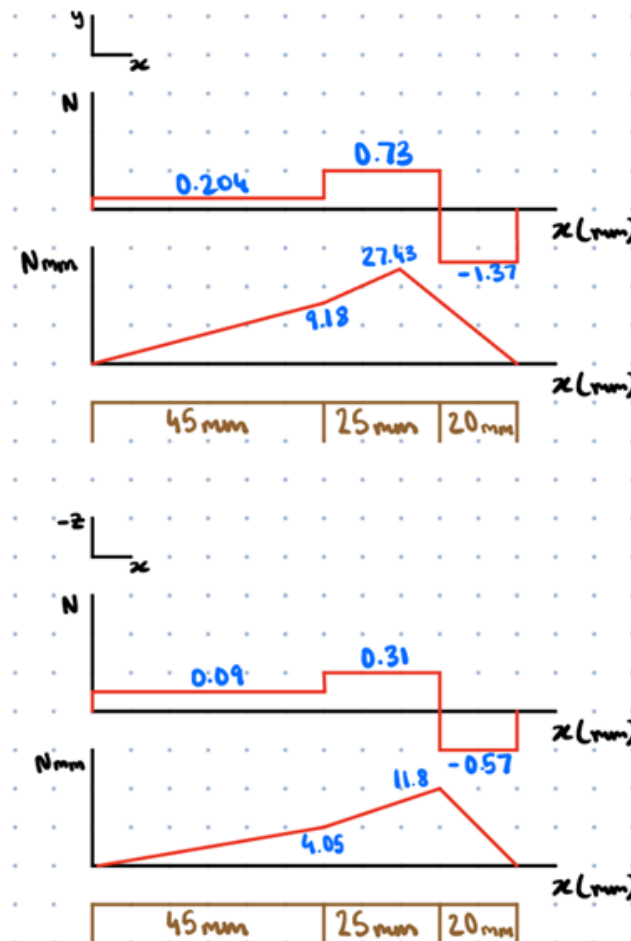
$$\sum F_y = 0 = B_{1y} + B_{2y} + F_{1t} - F_{2t} = B_{1y} + 1.37 + 0.526 - 2.1 \rightarrow B_{1y} = 0.204 \text{ N}$$

$$\sum F_x = 0 = -B_{2x} - F_{1a} + F_{2a} = -B_{2x} - 0.3 + 1.2 \rightarrow B_{2x} = 0.9 \text{ N}$$

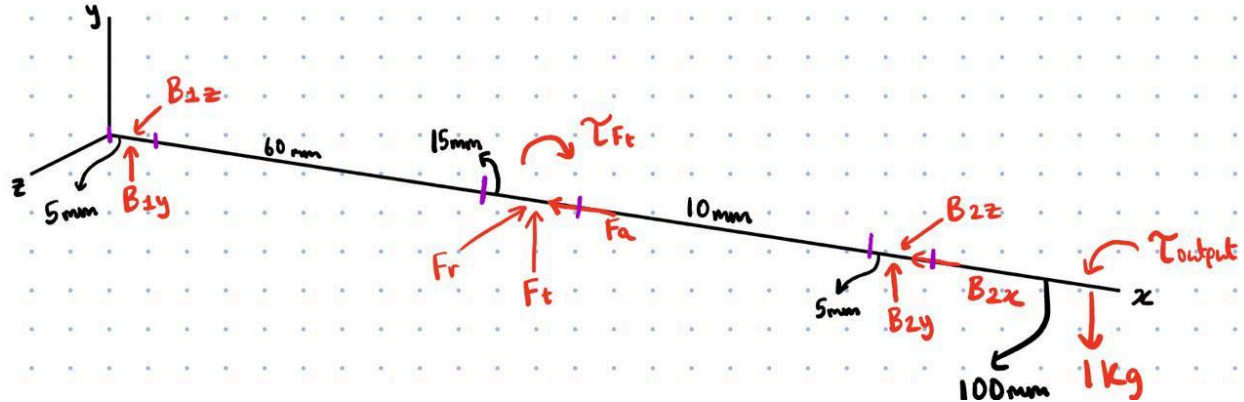
xz plane:

$$\begin{aligned} \sum M_{B1} = 0 = F_{1r} \left(\frac{5}{2} + 35 + \frac{15}{2} \right) - F_{2r} \left(\frac{5}{2} + 35 + 15 + 10 + \frac{15}{2} \right) - B_{2z} \left(\frac{5}{2} + 35 + 15 + 10 \right. \\ \left. + 15 + 10 + \frac{5}{2} \right) = 0.22(45) - 0.88(70) - B_{2z}(90) \rightarrow B_{2z} = -0.57 \text{ N} \end{aligned}$$

$$\sum F_z = 0 = B_{1z} + B_{2z} - F_{1r} + F_{2r} = B_{1z} - 0.57 - 0.22 + 0.88 \rightarrow B_{1z} = -0.09 \text{ N}$$



SHAFT 4



From shaft 3 calculations:

$$F_r = 0.88 \text{ N}$$

$$F_t = 2.1 \text{ N}$$

$$F_a = 1.2 \text{ N}$$

yz plane:

$$\begin{aligned} \sum M_O = 0 &= -\tau_{F_t} + \tau_{\text{output}} = -\frac{60(2.1)}{2\pi(11000)}(3 \times 4 \times 5) + \tau_{\text{output}} \\ &= -\frac{60(2.1)}{2\pi(11000)}(60) + \tau_{\text{output}} \rightarrow \tau_{\text{output}} = 0.11 \text{ Nm} \end{aligned}$$

xy plane:

$$\begin{aligned} \sum M_{B1} = 0 &= F_t \left(\frac{5}{2} + 60 + \frac{15}{2} \right) + B_{2y} \left(\frac{5}{2} + 60 + 15 + 10 + \frac{5}{2} \right) \\ &\quad - 9.81 \left(\frac{5}{2} + 60 + 15 + 10 + 5 + 100 \right) = 2.1(70) + B_{2y}(90) - 9.81(192.5) \\ &\rightarrow B_{2y} = 19.34 \text{ N} \end{aligned}$$

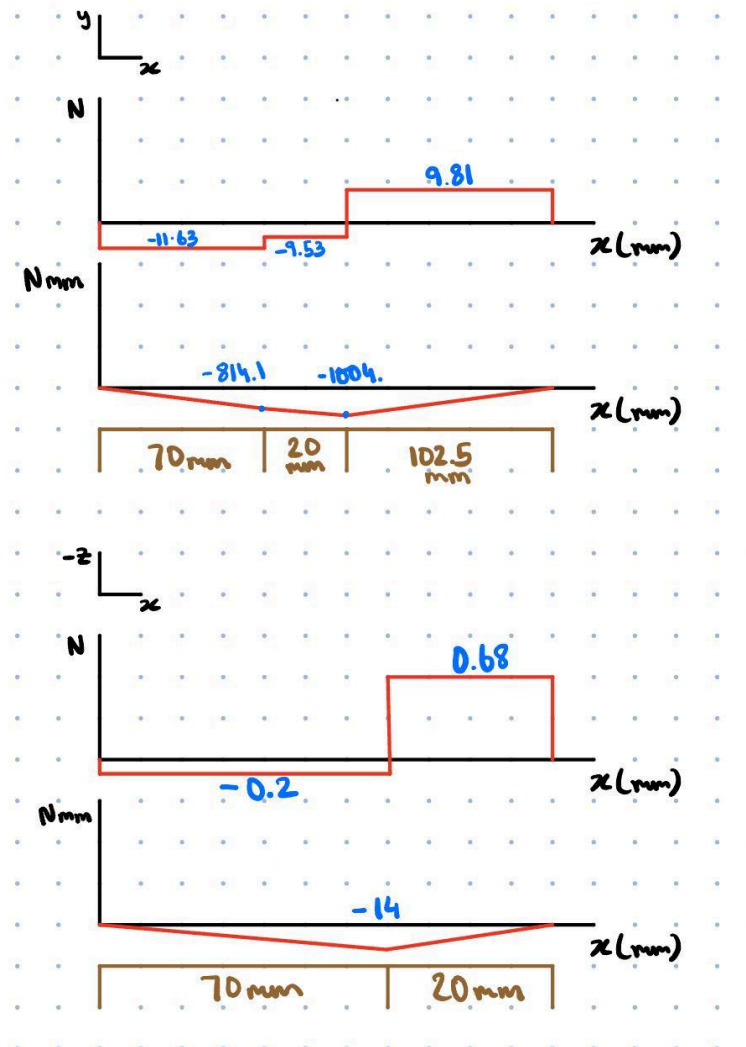
$$\sum F_y = 0 = B_{1y} + B_{2y} + F_t - 1 = B_{1y} + 19.34 + 2.1 - 9.81 \rightarrow B_{1y} = -11.63 \text{ N}$$

$$\sum F_x = 0 = -B_{2x} - F_a = -B_{2x} + 1.2 \rightarrow B_{2x} = -1.2 \text{ N}$$

xz plane:

$$\sum M_{B1} = 0 = F_r \left(\frac{5}{2} + 60 + \frac{15}{2} \right) - B_{2z} \left(\frac{5}{2} + 60 + 15 + 10 + \frac{5}{2} \right) = 0.88(70) - B_{2z}(90) \rightarrow B_{2z} = 0.68 \text{ N}$$

$$\sum F_z = 0 = B_{1z} + B_{2z} - F_r = B_{1z} + 0.68 - 0.88 \rightarrow B_{1z} = 0.2 \text{ N}$$



Stress analysis: For Critical Shaft

Von Mises Stress: Critical Shaft

1. Finding the Critical Cross-sections

$$M_b = \sqrt{M_y^2 + M_z^2}$$

At the Gear (x = 70mm)

$$M_z = -814.1 \text{ Nmm}, M_y = -14 \text{ Nmm}$$

Total Bending Moment

$$M_b = \sqrt{(-814.1)^2 + (-14)^2} = \mathbf{814.22 \text{ Nmm}}$$

At Bearing 2 (x = 90mm)

$$M_z = -1004.7 \text{ Nmm}$$

$$M_y = 0$$

$$M_b = \sqrt{(-1004.7)^2 + (0)^2} = \mathbf{1004.7 \text{ nmm}}$$

Bearing 2 (x = 90 mm) is the critical location as it sustains both the maximum internal torque (109.38 Nmm) and the peak bending moment (1004.7 Nmm).

2. Calculate Section Properties

$$A = \frac{\pi d^4}{4} = \frac{\pi 8^4}{4} = 50.27 \text{ mm}^2$$

$$I = \frac{\pi d^4}{64} = \frac{\pi 8^4}{64} = 201.06 \text{ mm}^4$$

$$J = \frac{\pi d^4}{32} = \frac{\pi 8^4}{32} = 402.12 \text{ mm}^4$$

3. Bending Stress

$$\sigma_b = \frac{M_b c}{I} = \frac{32 M_b}{\pi d^3} = \frac{32(1004.7)}{\pi(8)^3} = 19.99 \text{ MPa}$$

4. Axial Stress

$$\sigma_a = \frac{F_x}{A} = \frac{1.2}{50.27} = 0.024 \text{ MPa}$$

5. Total Normal Stress

$$\sigma_x = \sigma_b + \sigma_a = 19.99 + 0.024 = \mathbf{20.014\ MPa}$$

6. Torsional Shear Stress

$$\tau_{xy} = \frac{T \cdot r}{J} = \frac{16T}{\pi d^3} = \frac{16 \times 109.38}{\pi (8^3)} = \mathbf{1.09\ MPa}$$

6. Calculate the Von Mises Stress

$$\sigma_v = \sqrt{\sigma_x^2 + 3\tau_{xy}^2} = \sqrt{20.014^2 + 3(1.09)^2}$$

$$\sigma_v = \mathbf{20.10\ MPa}$$

$\sigma_y = \sigma_z = 0, \tau_{yz} = \tau_{zx} = 0$, hence why the formula is shortened

Distortion Energy Theory

Material Name	Youngs Modulus (GPa)	Yield Strength (MPa)	Tensile Strength (MPa)
Aluminium 6063-O	68	49	110
PLA (100% infill)	1	35	35

$$n = \frac{S_y}{\sigma_v}, S_y = 49\ MPa$$

$$n = \frac{49}{20.1} = \mathbf{2.44}$$

The factor of safety falls well within a reasonable range of 1.5 - 2.5. It is safely within the elastic region and will not yield.

Fatigue Analysis

$$S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot S'_e$$

$$k_a = a \cdot S_{ut}^b = (4.51)(110)^{-0.265} = 1.3$$

$$k_b = \left(\frac{d}{7.62}\right)^{-0.107} = \left(\frac{8}{7.62}\right)^{-0.107} = 0.995$$

$$k_c = k_d = k_e = 1$$

$$S_{ut} = 110\ MPa$$

$$S'_e = 0.5S_{ut} = 0.5 (110) = \mathbf{55\ MPa}$$

$$S_e = (1.3). (0.995). (1). (1). (1). (55)$$

$$S_e = \mathbf{71.14\ MPa}$$

Equivalent Alternating Stress and Mean Stress

$$\sigma'_a = \sqrt{\sigma_a^2 + 3\tau_a^2} = \sqrt{19.99^2 + 3(0)^2} = 19.99\ MPa$$

$$\sigma'_m = \sqrt{\sigma_{am}^2 + 3\tau_m^2} = \sqrt{0.000576 + 3(1.1881)} = 1.89\ MPa$$

Fatigue Safety Factor

$$\frac{1}{n_f} = \frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}}$$

$$\frac{1}{n_f} = \frac{19.99}{71.14} + \frac{1.89}{110}$$

$$n_f = 3.35$$

Bearing selection analysis

Critical Shaft is Shaft 4

$$F_r = 0.88\ N$$

$$F_t = 2.1\ N$$

$$F_a = 1.2\ N$$

$$B_{1y} = -11.63\ N$$

$$B_{1z} = 0.2\ N$$

$$B_{2y} = 19.34\ N$$

$$B_{2z} = 0.68\ N$$

Bearing 1 resultant radial load:

$$F_{r1} = \sqrt{B_{1y}^2 + B_{1z}^2} = \sqrt{11.63^2 + 0.2^2} = 11.63\ N$$

Bearing 2 resultant radial load:

$$F_{r2} = \sqrt{B_{2y}^2 + B_{2z}^2} = \sqrt{19.34^2 + 0.68^2} = 19.35 \text{ N}$$

$F_{r2} > F_{r1}$, therefore Bearing 2 is the critical bearing

Equivalent Dynamic Bearing Load

$$Fe = XVFr + YFa$$

$$V = 1$$

F_a/C_0	e	$F_a/(VF_r) \leq e$		$F_a/(VF_r) > e$	
		X_1	Y_1	X_2	Y_2
0.014*	0.19	1.00	0	0.56	2.30
0.021	0.21	1.00	0	0.56	2.15
0.028	0.22	1.00	0	0.56	1.99
0.042	0.24	1.00	0	0.56	1.85
0.056	0.26	1.00	0	0.56	1.71
0.070	0.27	1.00	0	0.56	1.63
0.084	0.28	1.00	0	0.56	1.55
0.110	0.30	1.00	0	0.56	1.45
0.17	0.34	1.00	0	0.56	1.31
0.28	0.38	1.00	0	0.56	1.15
0.42	0.42	1.00	0	0.56	1.04
0.56	0.44	1.00	0	0.56	1.00
*Use 0.014 if $F_a/C_0 < 0.014$.					

Table 1: Equivalent radial load factors for ball bearings

$C_0 = 1.37 \text{ kN}$ for 608 bearing

$$\frac{F_a}{C_0} = \frac{0.00121}{1.37} = 0.000883 < 0.014$$

$$\frac{F_a}{C_0} = 0.014 \text{ and } e = 0.19$$

$$\frac{F_a}{VF_r} = \frac{1.21}{1(1.766)} = 0.685$$

$$0.685 > 0.19 \text{ Therefore, } \frac{F_a}{VF_r} > e$$

$$X_2 = X = 0.56$$

$$Y_2 = Y = 2.3$$

$$F_e = 0.56(1)(1.766) + 2.3(1.21) = 3.78 \text{ N}$$

$$P = F_e = 3.78 \text{ N}$$

Required Bearing Life

$$15kh = 15000h$$

$$n_4 = \frac{n_{input}}{i_{total}} = \frac{11000 \text{ rpm}}{5(4)(3)} = 183.33 \text{ rpm}$$

$$L_n = \frac{15000(183.33)(60)}{10^6} = 165 \text{ million revolutions}$$

For 98% reliability:

$$a_1 = 0.37$$

For ball bearings:

$$p = 3$$

Bearing life equation:

$$L_n = a_1 \left(\frac{C}{P} \right)^3$$

$$C = P \left(\frac{L_n}{a_1} \right)^{\frac{1}{3}} = 3.78 \left(\frac{165}{0.37} \right)^{\frac{1}{3}} = 28.879 \text{ N}$$

$$C = 0.028879 \text{ kN}$$

$$C_{required} \geq 0.028879 \text{ kN}$$

Selected Bearing

608 deep-groove ball bearing

$$C_{selected} = 3.45 \text{ kN}$$

$$C_{required} = 0.0289 \text{ kN}$$

Since $3.45 \text{ kN} > 0.0289 \text{ kN}$, the 608 bearing is safe for the required life.

Life Expectancy of Selected Bearing

$$L_n = a_1 \left(\frac{C}{P} \right)^3$$

$$C = 3.45 \text{ kN}$$

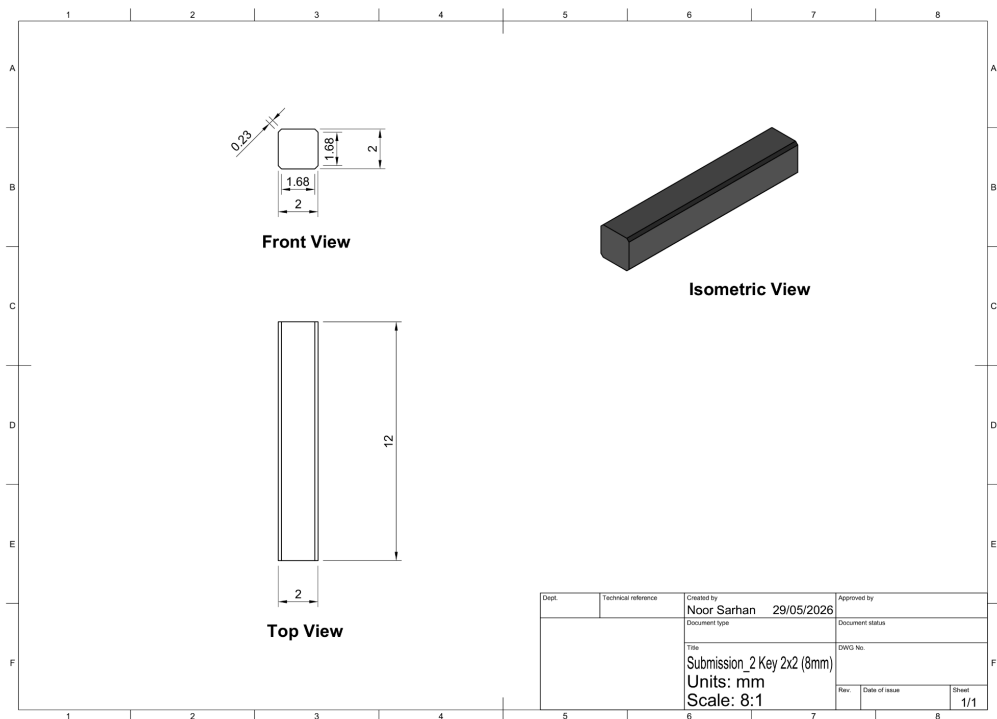
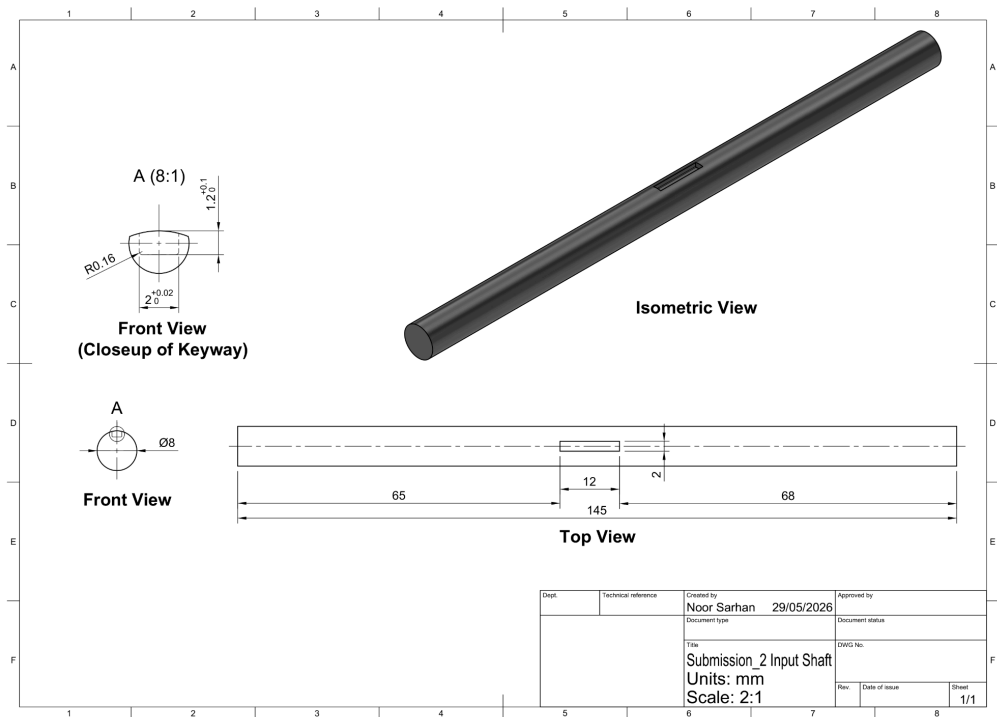
$$P = 3.78 \text{ N} = 0.00378 \text{ kN}$$

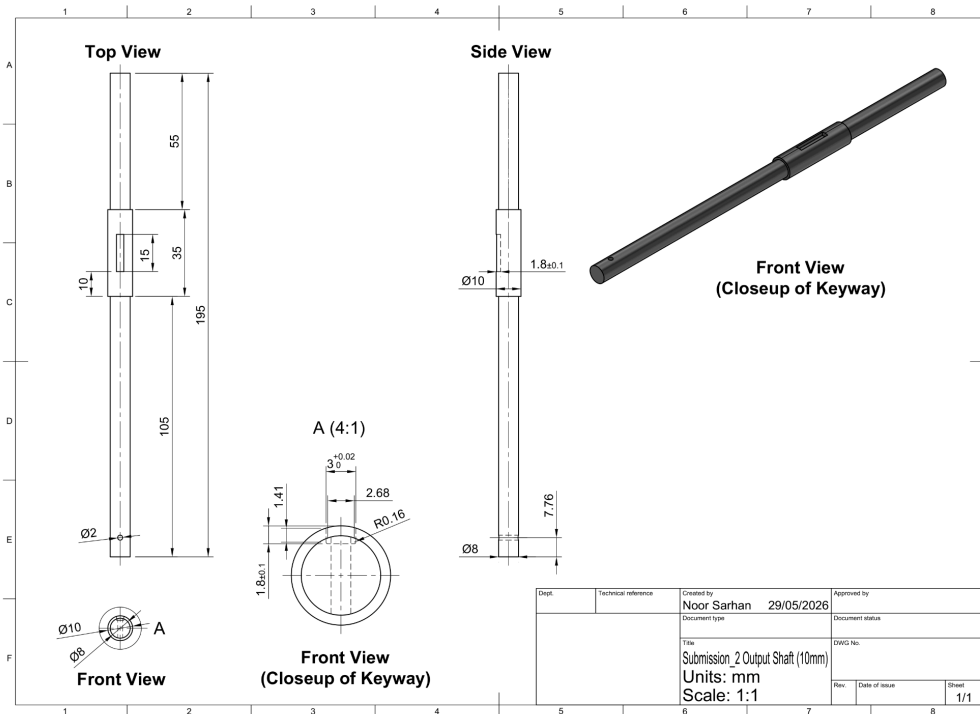
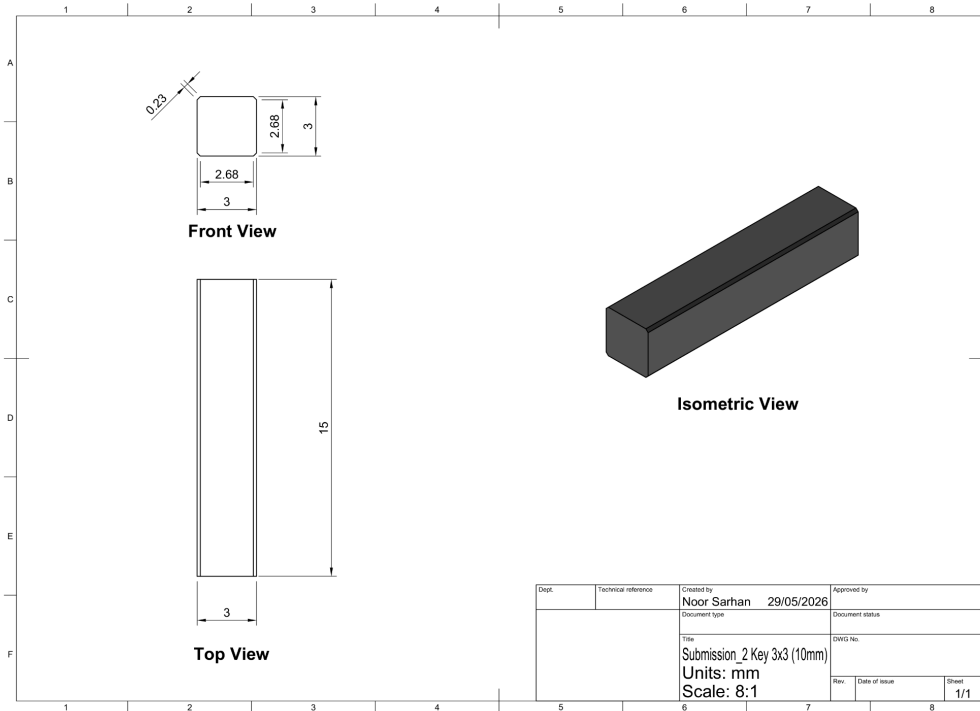
$$L_n = 0.37 \left(\frac{3.45}{0.00378} \right)^3 = 2.82 \times 10^8 \text{ million revolutions}$$

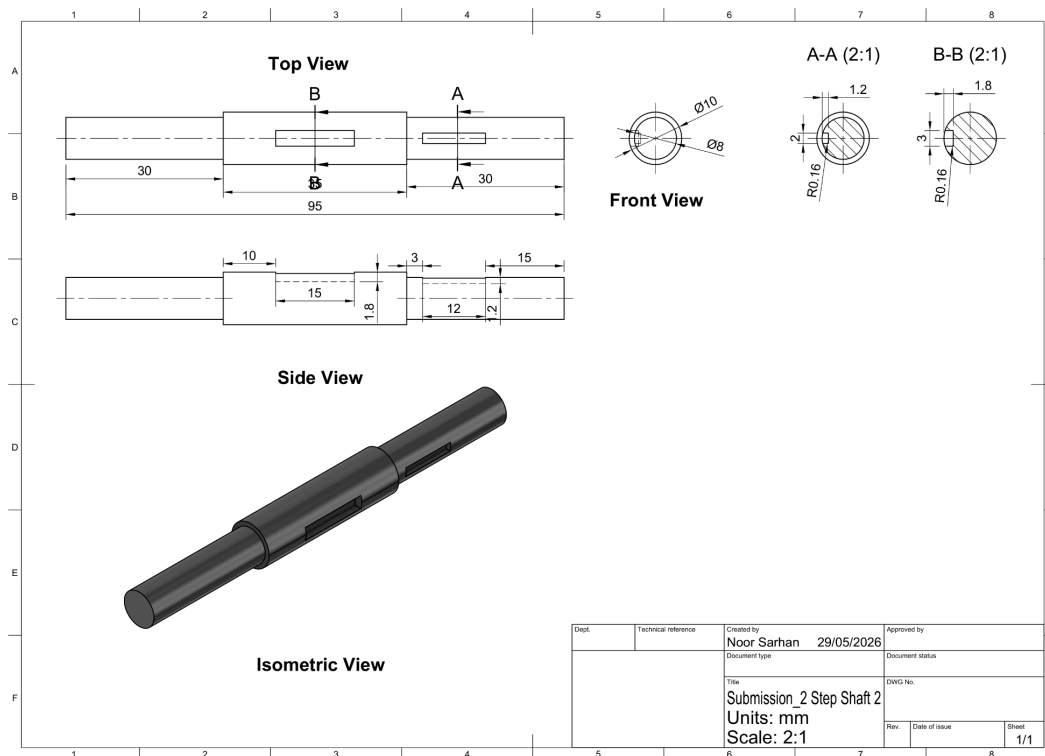
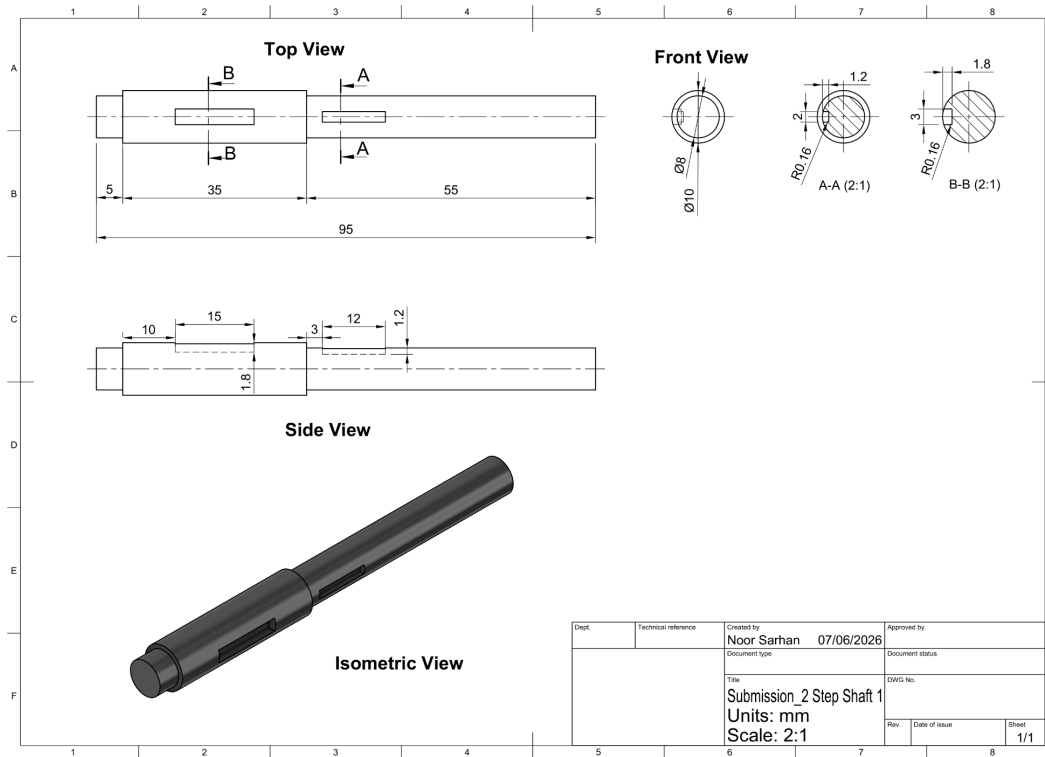
$$L_h = \frac{L_n (10^6)}{60n} = \frac{(2.82 \times 10^8)(10^6)}{60(183.33)} = 2.56 \times 10^{10} \text{ h}$$

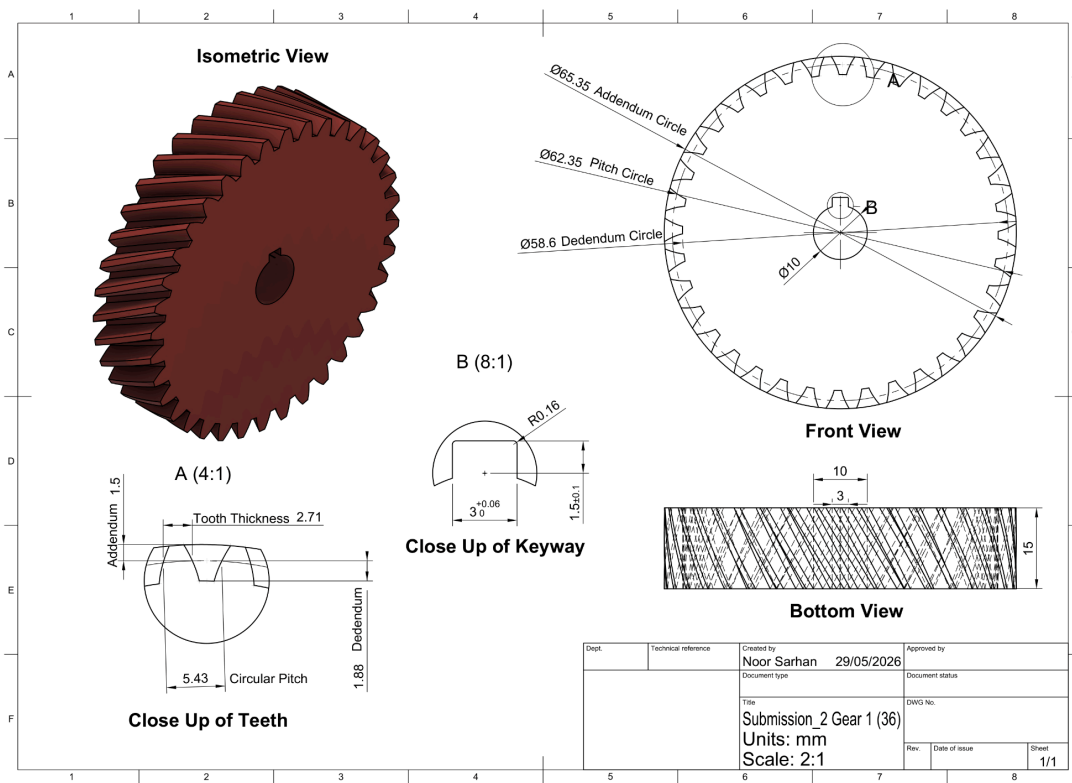
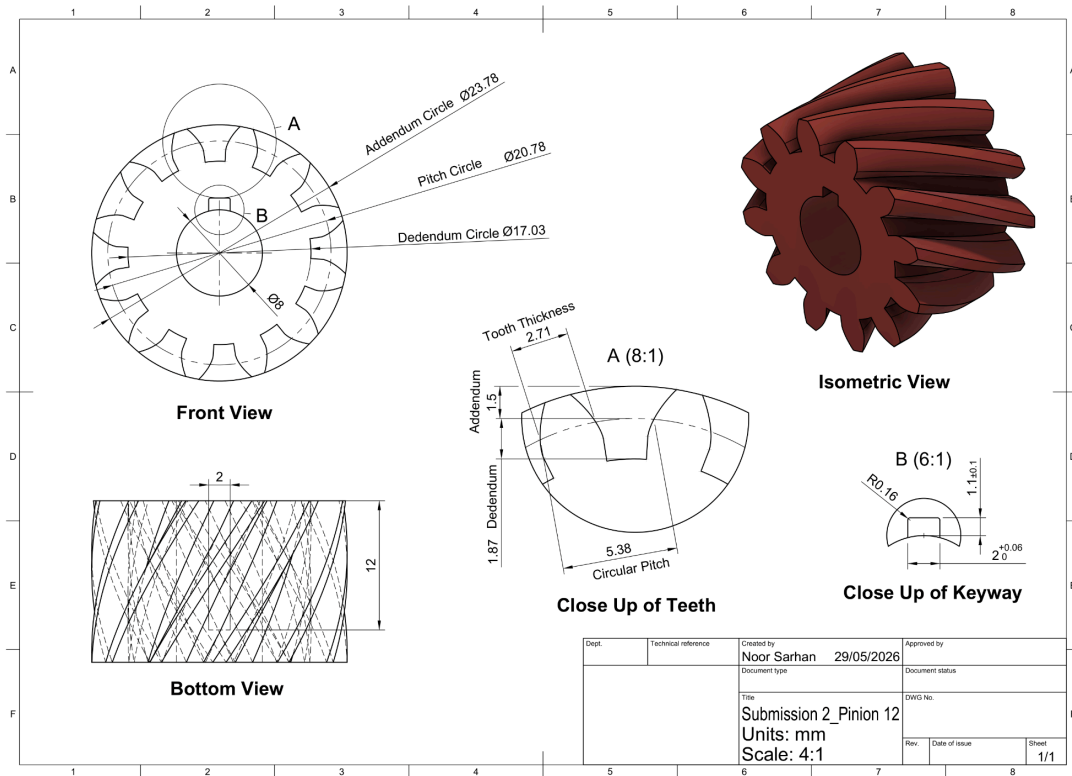
Since $2.56 \times 10^{10} \text{ h}$ is significantly greater than 15000 h, the selected 608 bearing will be more than sufficient for the gearbox.

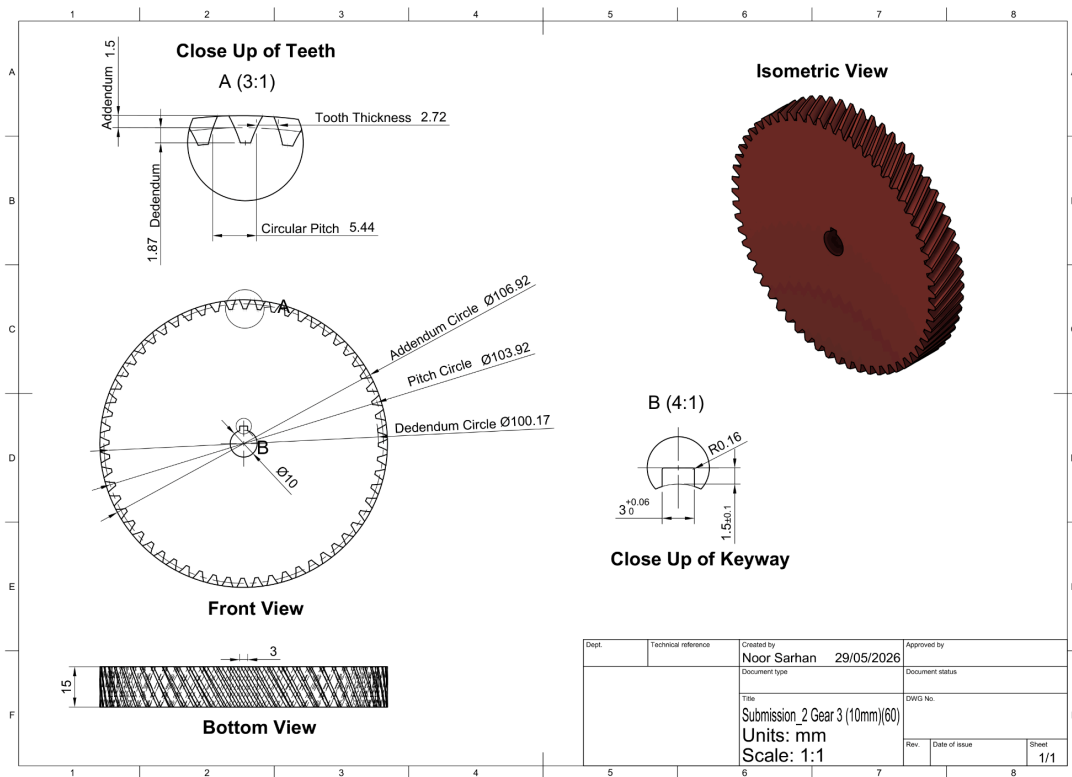
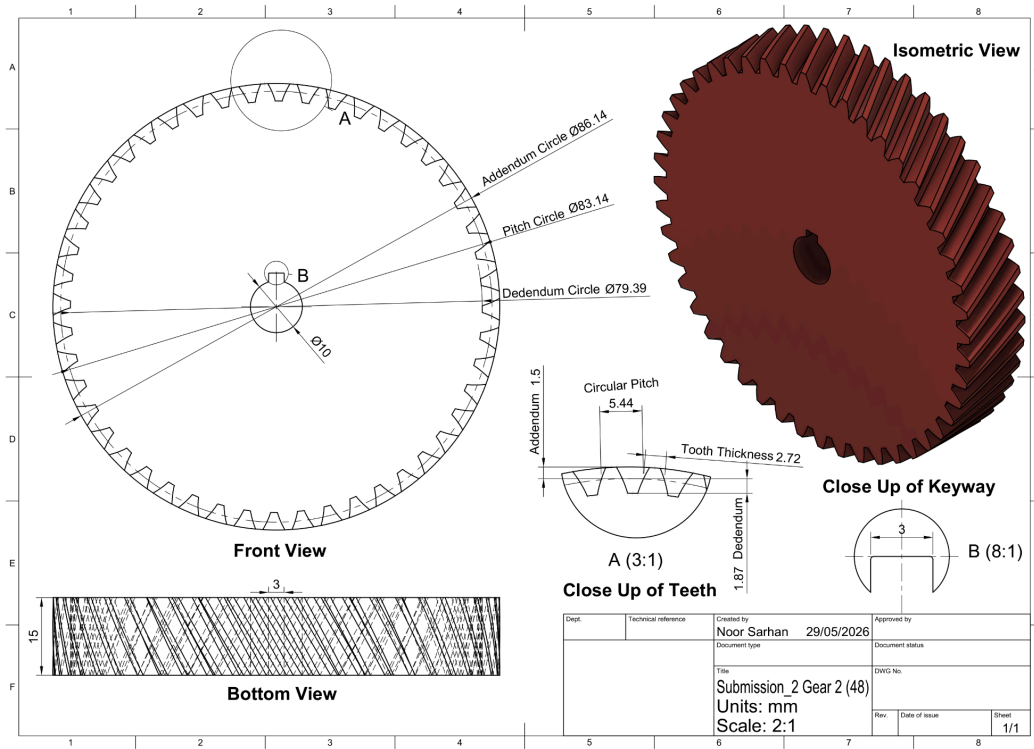
1.2. Task 2: Engineering/Assembly Drawings

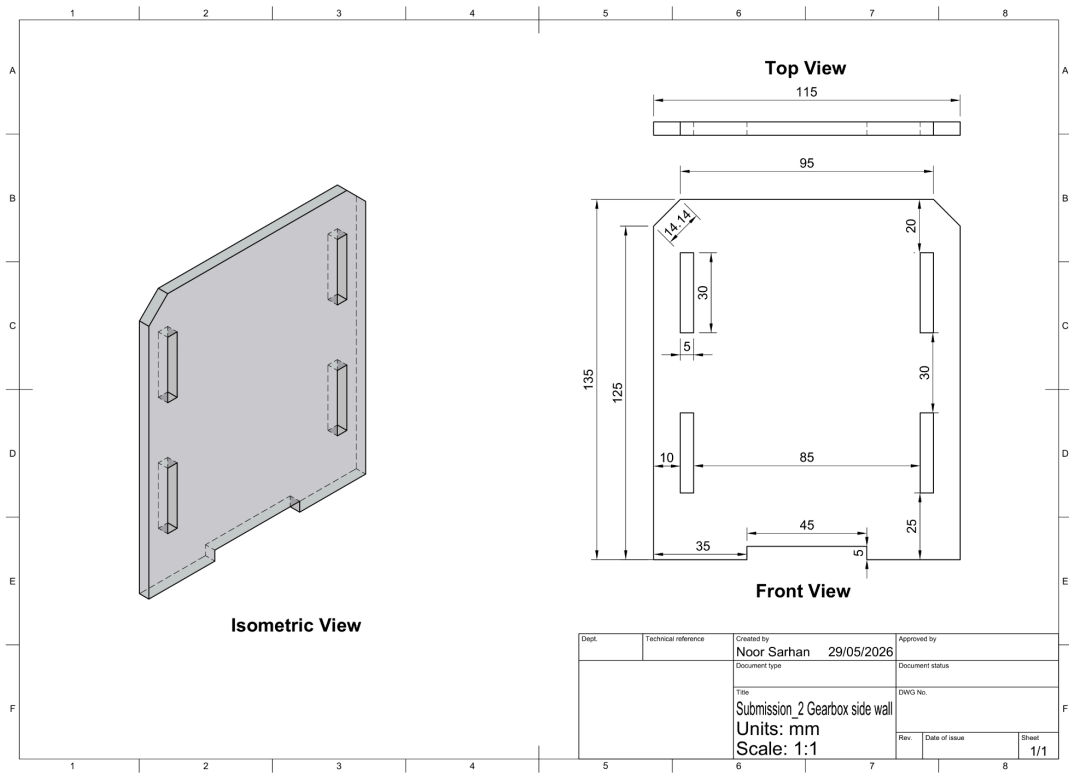
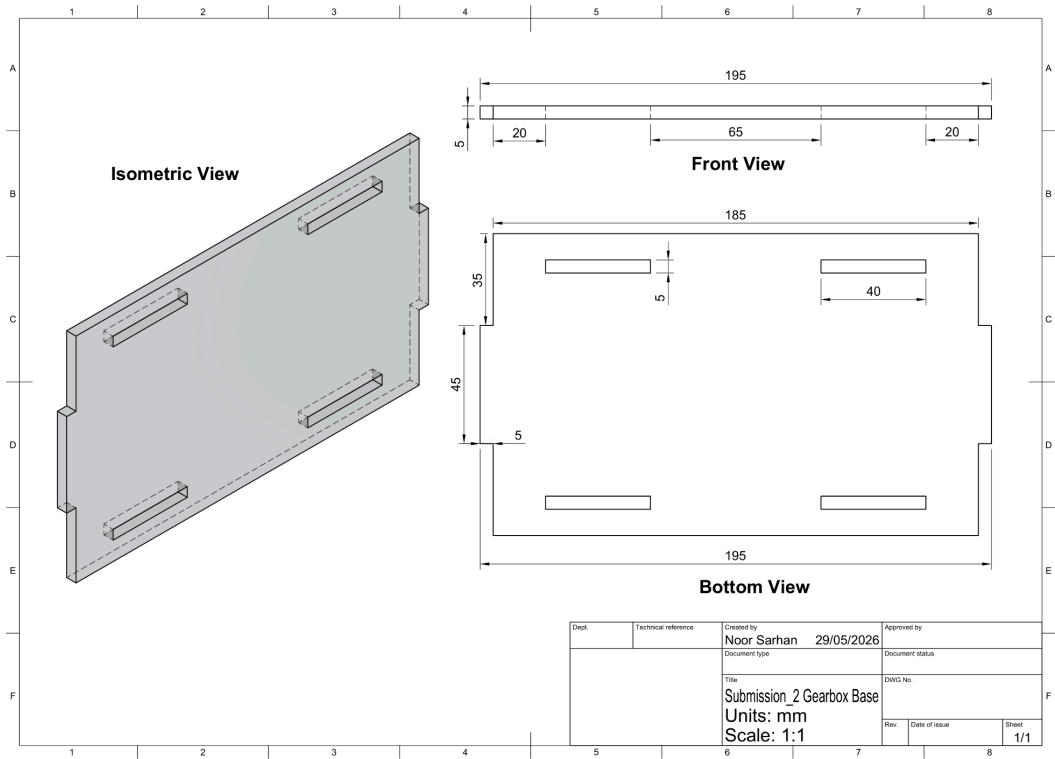


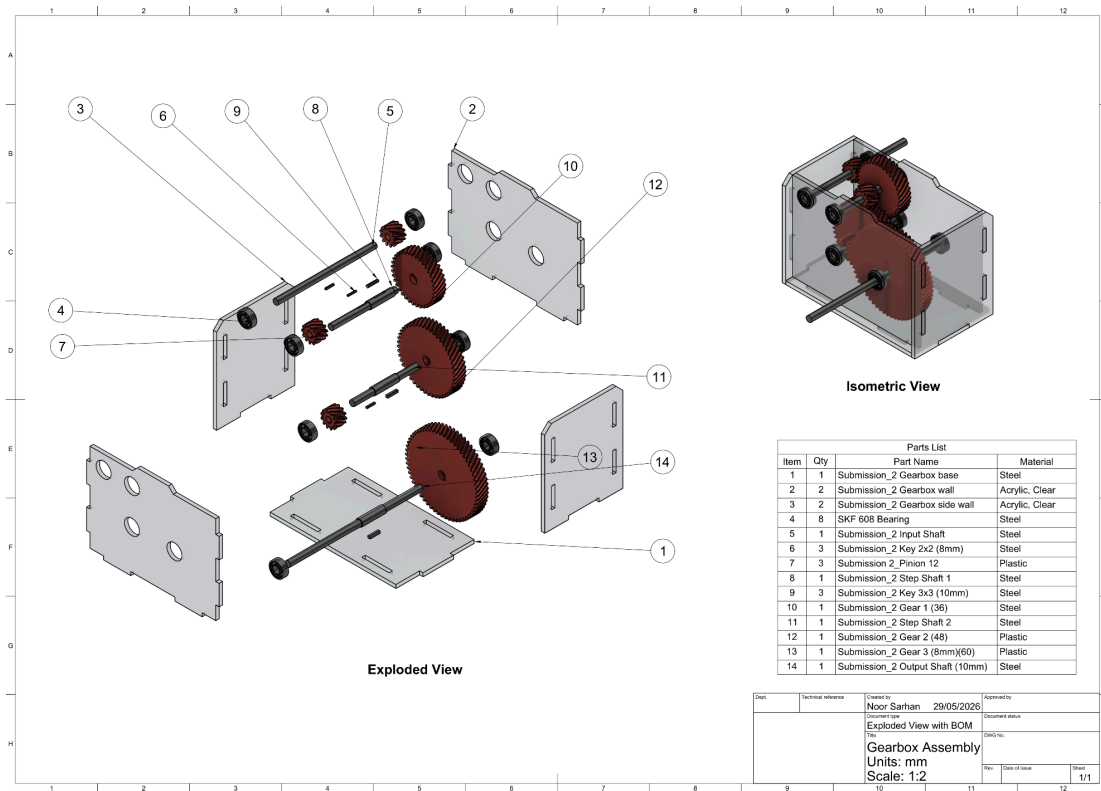
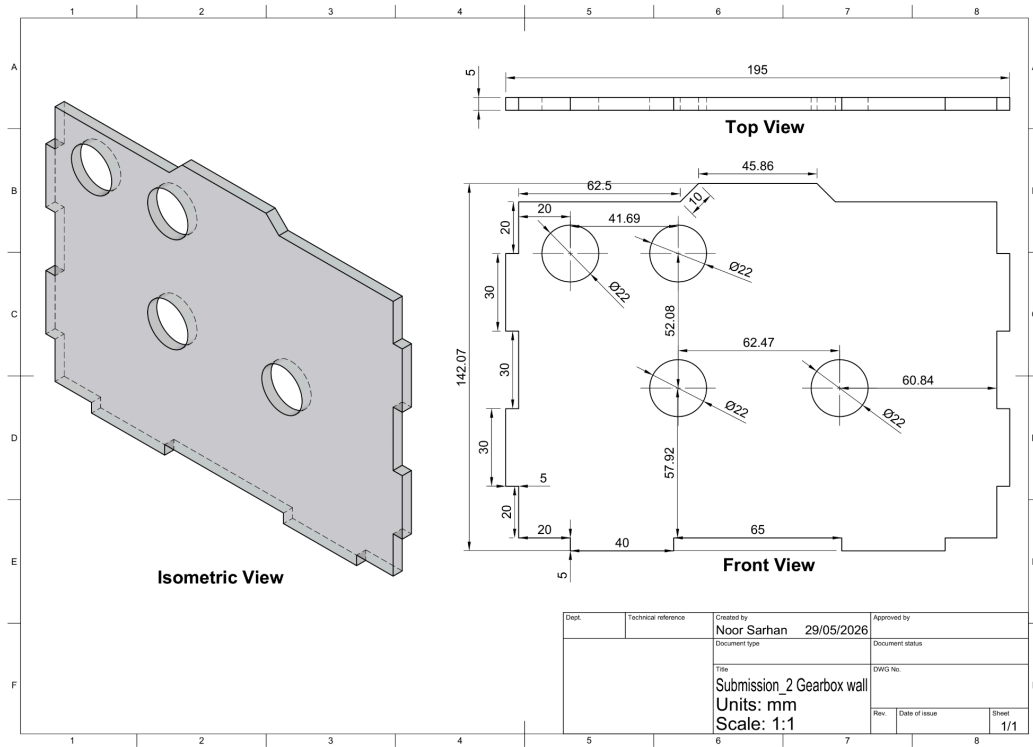


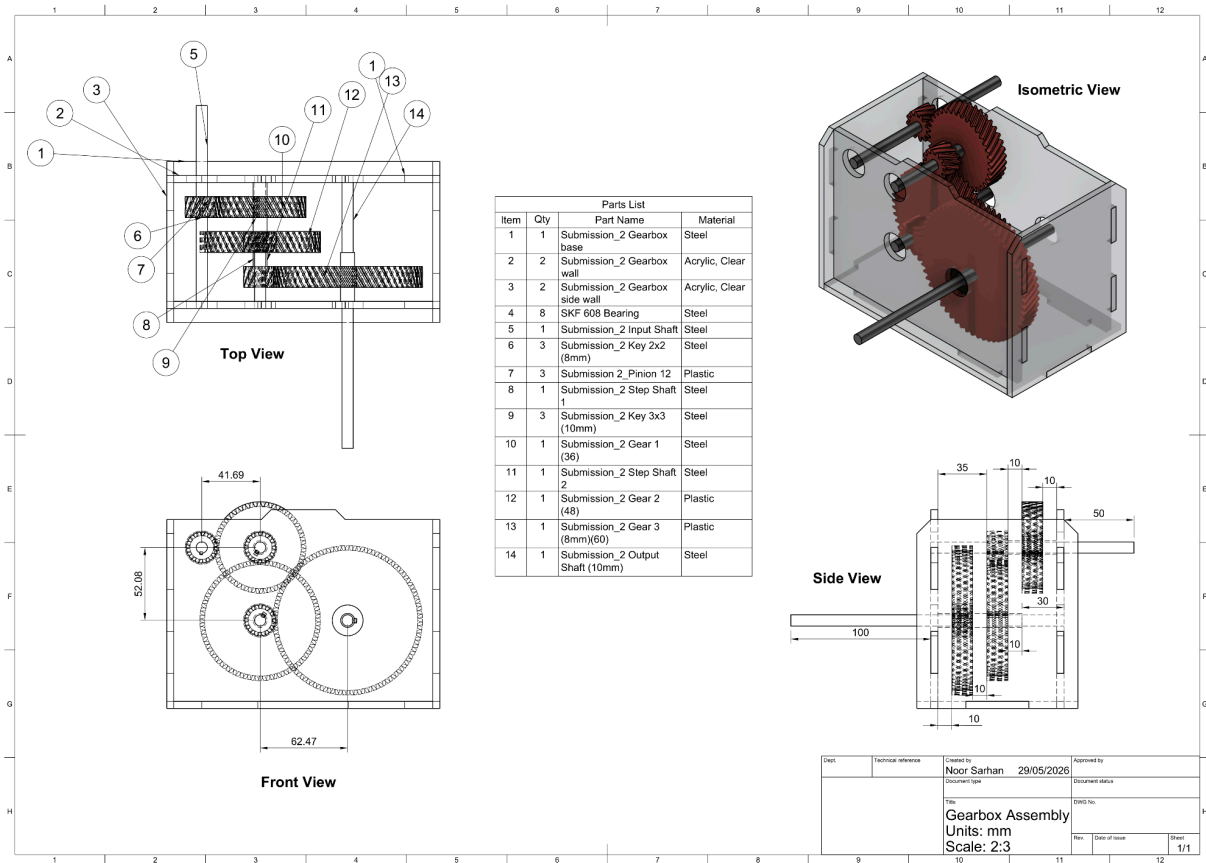












Summary/Next Steps

The gearbox design was sufficiently finalized. The number of teeth and arrangement of the gear ratios were changed in order to fit within the 200x200 mm constraint, hence the calculations were adjusted based on that. In addition, the layout of the gears was also changed for a more efficient setup.

The helical gear forces were successfully mapped and all support reactions were resolved. Thus, the critical shaft was located, in which the DE theory was used to find the factor of safety ($n = 2.44$), which demonstrates that the shaft is strong enough to handle the expected torque. As well as finding the fatigue factor of safety ($n_f = 3.35$). Since n_f is greater than 1, the shaft enters the 'infinite life' region, according to the Modified Goodman criteria. This means that theoretically the shaft is immune to fatigue failure.

The standard 608 ball bearing selection was verified against the resultant radial loads, and the expected operational life vastly exceeded the 15,000 hour target milestone.

As for the next steps, a machine will be used to carve the 10mm diameter aluminum shafts to create a step shaft, then a milling machine to carve keyseats for the keys. The gears will then be printed flat on the build plate, ensuring 100% infill as specified. Before permanently pressing the 608 bearings into the gearbox housing, a pre-assembly check will be performed by sliding the components into the output shaft outside of the casing. This will allow for the physical

verification of the helical gear-mesh alignment. Checking these tolerances will ensure that everything aligns smoothly before committing to final assembly.

Contributions Sheet

There have been a number of instances where particular group members have not engaged in the project to an adequate level (sometimes making no contribution).

To take account of this, each group is required to list the percentage contribution of each group member (totaling 100%).

Each group member must sign this sheet in acknowledgment that they accept the contributions listed against them.

In the case where there are differences in the level of contribution by group members, the final individual group mark will be adjusted up or down.

Group number/name:

Group Member Name	Percentage Contribution
John Patrick Traya	25%
Jude Khalil	25%
Noor Sarhan	25%
Shaik Ikramuddin Jeelani	25%